OPERATION KOMOREBI / REALIZATION CALCULUS

Paper III: Section 17

Packet jets, exact reconstruction and higher composition

Technical proof continuation 0.3.0 | 19 October 2026

This guide reports new ordinary mathematical arguments and reproducible exact checks. It does **not** certify two new Lean proof modules: the pinned compiler and mathlib cache were unavailable in this execution environment. The complete, previously checked Section 16 package is preserved unchanged.

The structural result

Let two finite families of linear maps resolve the identities on their respective modules. Any multilinear operator then admits an exact expansion over every input packet label and each output packet label. No symmetry assumption is required.

$$B(h) = \sum_{\gamma} \sum_{\alpha \in I^r} Q_\gamma B(P_{\alpha_1} h_1, \ldots, P_{\alpha_r} h_r)$$

A useful strengthening

The expansion requires only completeness of the linear families. The stronger idempotence and pairwise annihilation hypotheses belong to the assertion that packet coordinates are genuinely invertible. Keeping these statements separate prevents a decomposition identity from being mistaken for a coordinate certificate.

Current disposition

Complete written proof; 2,622 passed exact assertions and four rejected false shortcuts. Two source-level Lean candidates are preserved as uncompiled drafts. No new kernel-checked declarations are claimed.

1 Exact multilinear reconstruction

Suppose R is a semiring, X and Y are R-modules, and the index sets I and J are finite. Take linear operators P_i on X and Q_γ on Y satisfying the following identities.

$$\sum_{i \in I} P_i = \mathrm{id}_X, \qquad \sum_{\gamma \in J} Q_\gamma = \mathrm{id}_Y$$

For a multilinear map B with r arguments, insert the input identity independently into each argument and use additivity once per input slot. The r-fold finite expansion indexes all functions from the slot set to I. Insert the output identity and distribute it over the finite sum. This proves the exact equality on the previous page. No limiting operation is involved.

A literal jet with range codomain

Under the stronger projector hypotheses of Section 16, each input component lies in its actual projector range. Its output is in the range of Q_γ. Thus the jet is not merely a masked ambient expression but a well-defined multilinear map between actual range modules.

$$J_{\gamma \mid \alpha} B : \prod_{a=1}^{r} \operatorname{ran} P_{\alpha_a} \longrightarrow \operatorname{ran} Q_\gamma$$

The manuscript's derivative theorem follows by substituting $B = D^r F(x)$, where the needed differentiability hypotheses ensure that the Fréchet derivative is a continuous multilinear map. This correspondence is established mathematically, but its iterated derivative API binding is not yet Lean-checked.

Why the hypotheses matter

Over the field with two elements, take three copies of the identity as input projectors. Their sum is the identity and each is idempotent, yet they are not pairwise annihilating. The complete multilinear expansion still holds. However, analysis after synthesis sends (1, 0, 0) to (1, 1, 1). This is a concrete failure of unique packet coordinates without annihilation.

2 Normed packet jets

For analytic statements, the packet projectors must be bounded. Algebraic linearity alone does not guarantee this in infinite-dimensional Banach spaces. Take a continuous r-linear map B, bounded input projectors P_i and output projector Q_γ.

$$\|J_{\gamma \mid \alpha} B\| \leq \|Q_\gamma\| \, \|B\| \prod_{a=1}^{r} \|P_{\alpha_a}\|$$

Proof. Bound the image of B by its multilinear operator norm and the product of the input norms. Bound each projected input by the corresponding operator norm, then apply the output operator estimate. Supremizing over tuples in the product unit ball yields the claimed inequality. The genuine range-restricted jet inherits this bound from the ambient extension under the subspace norms.

The Lean norm draft uses mathlib's multilinear input-composition and output-composition inequalities. The draft has not been typechecked, and the bounded range-submodule construction remains an explicit formal obligation.

3 Composition at second order

The second-order chain rule has two distinct branches. For maps F after G, the outer Hessian applied to two first derivatives must be accompanied by the outer first derivative applied to the inner Hessian.

$$D^2(F \circ G)[h,k] = D^2F[DGh, DGk] + DF[D^2G[h,k]]$$

Insert the intermediate packet identity twice in the first branch and once in the second. A single intermediate label or diagonal-only restriction does not, in general, reproduce the derivative. The exact symbolic tests include 154 explicit cases detecting an invalid single-label shortcut.

4 Higher-order composition and counting

For general n, the classical higher-order chain rule is indexed by set partitions. A partition with k blocks receives one independent intermediate packet label on each block. With M possible intermediate packets, the number of decorated terms is the Touchard polynomial.

$$N_{n,M} = \sum_{k=1}^{n} S(n,k) M^k$$

This counting formula was checked by two independent constructions for n = 1 through 7 and M = 0 through 4: recursive set-partition enumeration and a Stirling-number recurrence. Such a count does not itself prove the full higher-order chain rule. Structural pruning removes summands with an identically zero constituent jet; cancellation may make the resulting support smaller, so only inclusion is generally justified.

5 Exact verification outcomes

Verified family	Number	Status
Multilinear reconstruction	75	PASS
Ambient jet norm inequality	174	PASS
Second-order decorated chain rule	192	PASS
Touchard labeled term counts	35	PASS
Stirling partition enumeration	28	PASS
Projector annihilation	660	PASS
All exact assertions (all families)	2622	PASS

The exact finite checks are evidence for small models, not a universal proof, Lean compilation, or computationally verified analytic continuation.

6 Release boundary and restart point

The only accepted Lean proof census remains the existing v0.2.0 Section 16 closure. The new Lean candidate files are deliberately isolated from the production import root. A static scan found no admitted, unsafe or opaque proof declarations, but a lexical scan cannot establish syntax, elaboration or kernel acceptance.

Compiler status

The current container does not have lean or lake on PATH. The pinned Lean 4.34.0-rc2 runtime described by the predecessor package is external to this delivery. Consequently there are zero new kernel-checked declarations in this v0.3.0 block.

Ordered next tasks

• Recover the pinned compiler and dependency objects and build both Section 17 draft files in an empty project object directory.

• For each accepted declaration, resolve type, ownership and transitive axioms; replay logical proof terms from upstream imports alone.

• Construct and prove the continuous normed-range and direct-sum bridge and the precise iterated Fréchet derivative correspondence.

• Formalize the projected Taylor formula, full finite-partition Faà di Bruno expansion, and support inclusion under their exact hypotheses.

• Reconcile the modern manuscript and Lean guide. Do not promote the unrelated October 16 98+2 / 2693 tally until its source and receipts are available.

Preservation

The full frozen predecessor, receipts, guides and historical sources are retained in the cumulative ZIP. Project licensing remains unresolved. Paper III must be finished before work proceeds chronologically to the next unfinished core paper.